

Gravitational Waves and Numerical Relativity: Example Sheet 1

Part III, Easter Term 2026

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1. Riemann tensor in linearized theory

Show that in linearized theory, the components of the Riemann tensor,

$$R_{\mu\nu\rho\sigma} = \frac{1}{2}(\partial_\rho\partial_\nu h_{\mu\sigma} + \partial_\sigma\partial_\mu h_{\nu\rho} - \partial_\rho\partial_\mu h_{\nu\sigma} - \partial_\sigma\partial_\nu h_{\mu\rho}),$$

are invariant under first-order coordinate transformations $\tilde{x}^\alpha = x^\alpha - \xi^\alpha$ where $\xi^\alpha = \mathcal{O}(\epsilon)$, $\epsilon \ll 1$.

2. Advection equation

Consider the *advection equation*

$$\partial_t f + \lambda \partial_x f = 0, \quad \lambda \in \mathbb{R},$$

for a function f on the domain $(t, x) \in \mathbb{R}^2$.

- (i) Show that f remains constant along curves $x(t) = \lambda t + x_0$, where $x_0 = \text{const.}$ These are the *characteristic curves* of the advection equation.
- (ii) Consider Gaussian initial data

$$f(0, x) = e^{-(x-x_0)^2}.$$

Sketch the characteristic curves in the (t, x) plane (with t pointing upwards and x horizontally). Also graphically sketch the solution $f(t, x)$. Quantitative precision is not required in this sketch and there are multiple ways to generate a graphic illustration; one figure is sufficient.

- (iii) Now consider the case of a varying $\lambda = \lambda(t, x)$ where we write the advection equation in the form

$$\frac{\partial f}{\partial t} + \frac{\partial}{\partial x}(\lambda(t, x)f) = 0.$$

The *characteristic curves* of this differential equation are solutions of

$$\frac{dx}{dt} = \lambda(t, x).$$

Show that the advection equation can be written as an ordinary differential equation along the characteristic curves.

3. Burgers' equation

A more dramatic variation of the advection equation arises when we allow the *flux* term λf to be a non-linear function of the evolution variable f , i.e. where λ itself depends on f . This results in so-called *quasi-linear* PDEs which are linear in the derivatives but not in the function f itself. The prototypical example for this type of PDEs is *Burgers' equation*, where $\lambda(f) = \frac{1}{2}f$. In a more common notation, this is written as $F := \lambda f = \frac{1}{2}f^2$, so that

$$\partial_t f + \partial_x F = \partial_t f + \partial_x \left(\frac{1}{2}f^2\right) = 0.$$

- (i) Determine the characteristic curves for this PDE, i.e. the curves along which the PDE can be written as an ordinary differential equation. Write down this ordinary differential equation.
- (ii) Sketch the characteristic curves in the (t, x) plane (with t pointing upwards and x horizontal) for Gaussian initial data

$$f(0, x) = e^{-(x-x_0)^2}.$$

- (iii) Sketch the time evolution of the Gaussian initial data. Compare the result with that obtained for the advection equation in Question 2.

4. Quasi-linear first-order PDE systems

Let $\mathbf{f} : \mathbb{R}^N \rightarrow \mathbb{R}^M$ and let us use the notation

$$\mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_M \end{pmatrix}, \quad \partial_\alpha \mathbf{f} := \frac{\partial}{\partial x_\alpha} \mathbf{f} = \begin{pmatrix} \partial_\alpha f_1 \\ \vdots \\ \partial_\alpha f_M \end{pmatrix}, \quad \alpha = 1, \dots, N.$$

The general *quasi-linear first-order PDE* for $\mathbf{f}$ is

$$\mathbf{A}^\mu(\mathbf{x}, \mathbf{f}) \partial_\mu \mathbf{f} + \mathbf{b}(\mathbf{x}, \mathbf{f}) = 0,$$

$$\Leftrightarrow A_{mn}^\mu(x_\alpha, f_i) \partial_\mu f_n + b_m(x_\alpha, f_i) = 0 \quad (\text{sum over } \mu, m, n), \quad (\dagger)$$

where $\mathbf{b}$ is a vector valued function, each $\mathbf{A}^\mu$ is an $M \times M$ matrix and initial data for $\mathbf{f}$ are given on a surface S defined as the level set of a function $\theta(x^\alpha) = 0$ with $\nabla \theta \neq 0$. We introduce coordinates adapted to this hypersurface S by

$$\xi^\alpha = \xi^\alpha(x^\mu) \quad \text{for } \alpha = 1, \dots, N-1,$$

$$\xi^N = \theta(x^\mu).$$

- (i) Show that the initial data $\mathbf{f}$ on S determine all derivatives $\partial f_i / \partial \xi_\alpha$ for $\alpha = 1, \dots, N-1$. Show that the PDE $(\dagger)$ also determines the derivative $\partial f_i / \partial \xi_N$ if and only if

$$\det \left(\mathbf{A}^\mu \frac{\partial \theta}{\partial x_\mu} \right) \neq 0.$$

- (ii) The *characteristic equation* associated with the PDE $(\dagger)$ is

$$\det \left(\mathbf{A}^\mu \frac{\partial \theta}{\partial x_\mu} \right) = 0, \quad (*)$$

and a level surface S defined through $\theta(x_\alpha) = 0$ by a solution to this equation is a *characteristic surface*.

- (a) Determine the matrices $\mathbf{A}^\alpha$ for the advection equation $\partial_t f + \partial_x f = 0$ and write down the characteristic equation. Note that the $\mathbf{A}^\alpha$ are just scalars in this case.
- (b) Introducing $\psi := \partial_t f$ and $\lambda := \partial_x f$, write the 1+1 dimensional wave equation $\partial_t^2 f - c^2 \partial_x^2 f = 0$, $c > 0$ as a first-order system of two equations, one each for $\partial_t \psi$ and $\partial_t \lambda$. [Hint: *Partial derivatives commute.*] Determine the $\mathbf{A}^\alpha$ and the characteristic equation.
- (c) Introducing $\psi := \partial_x f$ and $\lambda := \partial_y f$, write the 2 dimensional Laplace equation $\partial_x^2 f + \partial_y^2 f = 0$ as a first-order system of two equations, one each for $\partial_x \psi$ and $\partial_x \lambda$. Determine the $\mathbf{A}^\alpha$ and the characteristic equation.

5. Classification of first-order PDE systems

(This question directly proceeds from question 4 and employs all definitions therefrom.)

For the classification of first-order PDE systems, we define for an N dimensional vector ζ ,

$$\mathcal{C}(\mathbf{x}, \mathbf{f}, \zeta) := \det [\mathbf{A}^\mu(\mathbf{x}, \mathbf{f}) \zeta_\mu] .$$

We furthermore introduce a linear mapping

$$\zeta = \mathbf{M}\eta, \quad \text{where } \eta = (\eta_1, \dots, \eta_{N-1}, \kappa),$$

with a non-degenerate matrix $\mathbf{M}$, and write

$$\mathcal{C}(\mathbf{x}, \mathbf{f}, \eta) = \mathcal{C}(\mathbf{x}, \mathbf{f}, \eta_1, \dots, \eta_{N-1}, \kappa) := \mathcal{C}(\mathbf{x}, \mathbf{f}, \zeta(\eta_1, \dots, \eta_{N-1}, \kappa)) .$$

This reparametrization of the vector ζ is necessary to single out a specific parameter κ that does not necessarily coincide with a single component of ζ in the coordinates x_μ .

We then define the PDE

$$\mathbf{A}^\mu(\mathbf{x}, \mathbf{f}) \partial_\mu \mathbf{f} + b(\mathbf{x}, \mathbf{f}) = 0$$

to be

- *hyperbolic* at $\mathbf{x}$ if there exists a regular linear mapping $\zeta = \mathbf{M}\eta$, such that there exist M real roots $\kappa_i = \kappa_i(\mathbf{x}, \mathbf{f}(\mathbf{x}), \eta_1, \dots, \eta_{N-1})$, $i = 1, \dots, M$ of $\mathcal{C}(\mathbf{x}, \mathbf{f}, \eta) = 0$ for all $(\eta_1, \dots, \eta_{N-1})$. Note that the number of roots required equals the number of independent variables in $\mathbf{f}$, not the dimensionality N of the domain.
- *parabolic* at $\mathbf{x}$ if there exists a linear mapping $\zeta = \mathbf{M}\eta$ such that $\mathcal{C}$ is independent of κ , i.e. depends on fewer than N parameters.
- *elliptic* if $\mathcal{C}(\mathbf{x}, \mathbf{f}, \zeta) = 0$ only if $\zeta = 0$.

Determine the type of the advection equation, the 2-dimensional wave equation and 2-dimensional Laplace equation from question 4 according to this classification.

6. The constraint equations

Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime with a foliation Σ_t . Let $x^\alpha = (t, x^i)$ denote coordinates adapted to this foliation.

(i) Show that the components of the Ricci tensor can be written as

$$R_{00} = -\frac{1}{2} g^{mn} \partial_0^2 g_{mn} + M_{00},$$

$$R_{0i} = \frac{1}{2} g^{0m} \partial_0^2 g_{im} + M_{0i},$$

$$R_{ij} = -\frac{1}{2} g^{00} \partial_0^2 g_{ij} + M_{ij},$$

where M_{00} , M_{0i} and M_{ij} are a collection of terms that include at most first time derivatives of the metric $g_{\alpha\beta}$.

(ii) Show that the components $G_\alpha{}^0$ of the Einstein tensor can be written as

$$G_0{}^0 = \frac{1}{2} g^{00} M_{00} - \frac{1}{2} g^{mn} M_{mn},$$

$$G_i{}^0 = g^{00} M_{0i} + g^{0m} M_{im},$$

and therefore contain no second time derivative of the metric.

7. The Bondi metric

The Bondi metric is given by

$$g_{\alpha\beta} = \begin{pmatrix} -\frac{V}{r}e^{2\beta} + r^2U^2e^{2\gamma} & -e^{2\beta} & -r^2Ue^{2\gamma} & 0 \\ -e^{2\beta} & 0 & 0 & 0 \\ -r^2Ue^{2\gamma} & 0 & r^2e^{2\gamma} & 0 \\ 0 & 0 & 0 & r^2e^{-2\gamma}\sin^2\theta \end{pmatrix}. \quad (1)$$

Use the cofactor matrices to compute the inverse Bondi metric $g^{\alpha\beta}$.

8. Bianchi identities

Let $\mathcal{M}$ be a Lorentzian manifold with metric $g_{\alpha\beta}$. Show that the contracted Bianchi identities $\nabla^\mu G_{\alpha\mu}$ can be written as

$$g^{\mu\rho} \left(\partial_\rho R_{\alpha\mu} - \Gamma_{\mu\rho}^\sigma R_{\alpha\sigma} - \frac{1}{2} \partial_\alpha R_{\mu\rho} \right).$$